

data 05.12.25

S T Q Q S S D

Prova 01 Cálculo Numérico, Keurrg Hemyllii

Questão 01:

$$D=8$$

$$D_{n-1}=1$$

$$C=0,45 + \lfloor (1+8)/10 \rfloor$$

$$C=0,45 + \lfloor 9/10 \rfloor$$

$$C=0,45+0,9$$

$$C=1,35$$

$$F(\text{mid}) = 4,98 \cdot (0,25) - e^{0,25}$$

$$= 1,245 - 1,28402 = -0,0390$$

$$F(a) < 0, F(m) < 0$$

$$[a, b] = [0,25, 0,5]$$

$$\text{III} = a = 0,25 \quad b = 0,5$$

$$m = \frac{0,25 + 0,5}{2} = 0,3750$$

Questão 02:

$$F(x) = (4,98)x - e^x$$

$$5 < 0,0075$$

$$\text{I) } a = -0,5, \quad b = 0,5$$

$$m = \frac{-0,5 + 0,5}{2} = 0,0000 \text{ mid}$$

$$F(a) = 4,98(-0,5) - e^{-0,5}$$

$$= -2,49 - 0,60653 \approx -3,0953$$

$$F(a) = -0,0390$$

$$F(b) = 0,8412$$

$$F(\text{mid}) = 4,98 \cdot (0,375) - e^{0,375}$$

$$= 1,8675 - 1,45499 = 0,4125$$

$$F(a) < 0 \quad F(m) > 0$$

$$[a, b] = [0,25, 0,375]$$

$$\text{IV} = a = 0,25, \quad b = 0,375$$

$$m = \frac{0,25 + 0,375}{2} = 0,3125$$

$$F(b) = 4,98 \cdot 0,5 - e^{0,5}$$

$$= 2,49 - 1,64872 \approx 0,8412$$

$$F(a) = 0,0390$$

$$F(b) = 0,8412$$

$$F(m) = 4,98 \cdot (0,3125) - e^{0,3125}$$

$$= 1,55625 - 1,36683 = 0,1894$$

$$F(a) < 0 \quad F(b) > 0 \quad F(\text{mid}) < 0$$

$$[a, b] = [0,0, 0,5]$$

$$F(a) < 0 \quad F(m) > 0$$

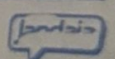
$$[a, b] = [0,25, 0,3125]$$

$$\text{II) } a = 0,0, \quad b = 0,5$$

$$m = (0 + 0,5) / 2 = 0,2500$$

$$F(a) = -1,0000$$

$$F(b) = 0,8412$$



$$\text{V) } a=0,25, b=0,3125$$

$$m = \frac{0,25 + 0,3125}{2} = 0,2813$$

$$F(a) = -0,0390$$

$$F(b) = 0,1894$$

$$F(m) = 4,98(0,2813) - e^{0,2813} \\ = 1,4006 - 1,3248 = 0,0758$$

$$F(a) \leq 0 \quad F(m) > 0$$

$$[a, b] = [0,25, 0,2813]$$

$$\text{VI) } a=0,25, b=0,2813$$

$$m = \frac{0,25 + 0,2813}{2} = 0,2656$$

$$F(a) = -0,0390$$

$$F(b) = 0,0758$$

$$F(m) = 4,98(0,2656) - e^{0,2656} \\ = 1,3239 - 1,3049 = 0,0190$$

$$F(a) \leq 0 \quad F(m) > 0$$

$$[a, b] = [0,25, 0,2656]$$

$$\text{VII) } a=0,25, b=0,2656$$

$$m = \frac{0,25 + 0,2656}{2} = 0,2578$$

$$F(a) = -0,0390$$

$$F(b) = 0,0189$$

$$F(m) = 4,98(0,2578) - e^{0,2578} \\ = 1,2810 - 1,2912 = -0,0102$$

$$F(a) < 0, F(m) < 0$$

$$[a, b] = [0,2578, 0,2656]$$

$$\text{VIII) } a=0,2578, b=0,2656$$

$$m = \frac{0,2578 + 0,2656}{2} = 0,2617$$

$$F(a) = -0,0102$$

$$F(b) = 0,0180$$

$$F(m) = 4,98(0,2617) - e^{0,2617} \\ = 1,3035 - 1,2993 = 0,0042$$

$$|F(m)| = 0,0042 < \epsilon = (0,0075)$$

Interações realizadas = 8

$$x = 0,2617$$

Valor da função no ponto: $f(0,2617)$

$$\approx 0,0042 = |F| < \epsilon$$

Questão 03:

$$F(x) = e^x + 0,21x - (3,49 + 1,35)$$

$$F(x) = e^x + 0,21x - 4,84$$

$$F'(x) = e^x + 0,21$$

$$\varepsilon = 0,000040$$

$$1 \rightarrow 2) \quad x_1 = 1,509233$$

$$F(x_1) = e^{1,509233} + 0,21(1,509233) - 4,84$$

$$= 4,523248 + 0,31693893 - 4,84$$

$$= 0,000187$$

$$F'(x_1) = e^{1,509233} + 0,21$$

$$= 4,523248 + 0,21 = 4,733298$$

$$F(1) = e^1 + 0,21(1) - 4,84 = -1,911718$$

$$F(2) = e^2 + 0,21(2) - 4,84 = 2,969056$$

$$\text{Portanto } [a, b] = [1, 2]$$

$$\Delta_1 = \frac{0,000187}{4,733298} = 0,0000395$$

$$x_2 = x_1 - \Delta_1 = 1,509233 - 0,0000395$$

$$\approx 1,509194$$

$$x_0 = \frac{1+2}{2} = 1,500000$$

critério I):

$$x_{k+1} = x_k - \frac{F(x_k)}{F'(x_k)}$$

$$\frac{|x_2 - x_1|}{2} = \frac{|1,509194 - 1,509233|}{2}$$

$$0 \rightarrow 1)$$

$$F(x_0) = e^{1,5} + 0,21(1,5) - 4,84$$

$$= 4,481689 + 0,315000 - 4,84$$

$$= 0,093311$$

$$= \frac{0,000039}{2} = 0,0000195 < 0,000040 \quad \checkmark$$

$$\text{Critério II): } F(x_2) \approx F(x_1) - F'(x_1) \cdot \Delta_1$$

$$= |F(x_2)| < \varepsilon \quad \checkmark$$

$$F'(x_0) = 4,481689 + 0,21 = 4,691689$$

$$\Delta_0 = \frac{F(x_0)}{F'(x_0)} = \frac{0,093311}{4,691689} \approx 0,009233$$

$$x_1 = x_0 - \Delta_0 = 1,500000 + 0,009233$$

$$= 1,509233$$

Questão 04:

$$x = 0,731 \text{ e } x = 1,126$$

$$L_0(0,731) = 0,287661 \approx 0,9191$$

$$0,6864$$

$$y_0 L_0 \approx 1,1977 \times 0,9191 \approx 0,9810$$

$$x_0 = 0,9500, \quad f(x_0) = y_0 = 1,1977$$

$$x_1 = 1,1100, \quad f(x_1) = y_1 = 2,7083$$

$$x_2 = 1,4900, \quad f(x_2) = y_2 = 4,9914$$

$$L_1 = (0,731 - 0,9500)(0,731 - 1,4900)$$

$$= (0,281)(-0,759) = -0,21329$$

$$L_1(0,731) = -0,21329 \approx -0,8507$$

$$0,2506$$

$$y_1 L_1 = 2,7083 \times 0,8507 \approx 2,3029$$

$$P(x) = \sum_{j=0}^2 y_j L_j(x), \quad L(x) = \prod_{m \neq j} \frac{x - x_m}{x_j - x_m}$$

$$L_2 = (0,731 - 0,9500)(0,731 - 1,1100)$$

$$= (0,281)(-0,379) = -0,106499$$

$$L_2(0,731) = -0,106499 \approx -0,2698$$

$$0,3952$$

$$y_2 L_2 = 4,9914 \times (-0,2698) \approx -1,3451$$

Denominadores Fixos

$$x_0 - x_1 = 0,9500 - 1,1100 = -0,6600$$

$$x_0 - x_2 = 0,9500 - 1,4900 = -1,0400$$

$$\text{Denom}_0 = (x_0 - x_1)(x_0 - x_2) = (-0,66)(-1,04)$$

$$= 0,6864$$

$$P(0,731) = 0,9810 + 2,3029 + (-1,3451)$$

$$P = -1,4588$$

$$x_1 - x_0 = 1,1100 - 0,9500 = 0,6600$$

$$x_1 - x_2 = 1,1100 - 1,4900 = -0,3800$$

$$\text{Denom}_1 = (x_1 - x_0)(x_1 - x_2) = (0,66)(-0,38)$$

$$= -0,2508$$

$$x_2 - x_0 = 1,4900 - 0,9500 = 1,0400$$

$$x_2 - x_1 = 1,4900 - 1,1100 = 0,3800$$

$$\text{Denom}_2 = (x_2 - x_0)(x_2 - x_1) = 1,0400 \cdot 0,3800$$

$$= 0,3952$$

Avaliação em $x = 0,731$

$$L_0(0,731)$$

$$\text{num}_0 = (0,731 - 1,1100)(0,731 - 1,4900)$$

$$= (-0,379)(-0,759) = 0,287661$$

Avaliação em $x=1,125$

$L_0(1,125)$

$$\text{num}_0 = (1,125 - 1,1100)(1,125 - 1,9900) \\ = (0,015)(-0,365) = -0,005475$$

$$L_0 = \frac{-0,005475}{0,6865} \approx -0,0080$$

$$y_0|_0 = 1,1477 \times (-0,0080) = -0,0092$$

$L_1(1,125)$

$$\text{num}_1 = (1,125 - 0,9500)(1,125 - 1,9900) \\ = (0,675)(-0,365) = -0,246375$$

$$L_1 = \frac{-0,246375}{0,2608} \approx -0,9823$$

$$y_1|_1 = 2,7083 - 0,9823 \approx 2,6609$$

$L_2(1,125)$

$$\text{num}_2 = (1,125 - 0,9500)(1,125 - 1,1100) \\ = (0,675)(0,015) = 0,010125$$

$$L_2 = \frac{0,010125}{0,3952} \approx 0,0256$$

$$y_2|_2 = 49919 \cdot 0,0256 \approx 0,1280$$

$$P(1,125) = (-0,0092) + 2,6609 + 0,1280$$

$$P = 2,7797$$